PCB & MANUFACTURING SPECIFICATION · SKU PIA24027P

4-Layer Board — Fabrication & Assembly Data

Bare-board material, the full 4-layer stack-up, design rules, drill data and assembly figures for Octopus 16. Circuit diagrams, copper artwork and the parts list are not part of this document.



4	1.6 mm	Ø 26	0.15 mm	0 THT
Copper layers	FR-4 thickness	mm round outline	Minimum track	All surface mount

01 BARE-BOARD FABRICATION PARAMETERS

Parameter	Specification	Note
Base material	**FR-4**, εr 4.5, tan δ 0.02	Standard Tg
Layers	**4**	Signal / signal / signal / signal, with ground pour on top
Dimensions	**Ø 26.0 mm** (26 × 26 mm bounding box)	Single round board
Thickness	**1.6 mm**	See stack-up
Solder mask	**Black**, both sides	As production boards
Silkscreen	**White**	Reference designators, jumper names
Surface finish	**HASL** (production) · ENIG optional	ENIG gives flatter pads for the pogo pins
Copper weight	**1 oz** (35 µm) all layers	
Vias / RoHS	Tented both sides · lead-free	

02 LAYER STACK-UP

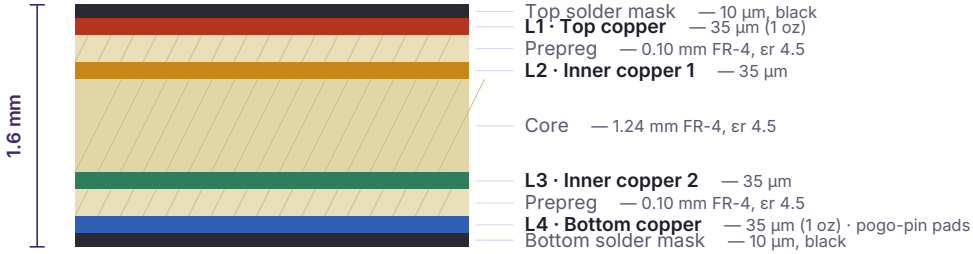


Figure 1 — Production stack-up. 0.10 mm prepreg | 1.24 mm core | 0.10 mm prepreg; 35 µm copper; 10 µm mask.

For research and education only. Not a medical device. Specifications subject to change.

03 DESIGN RULES (MEASURED FROM PRODUCTION DATA)

Feature	Value
Minimum track width	0.15 mm (6 mil)
Track widths used	0.15 · 0.2 · 0.3 · 0.4 · 0.5 · 0.8 mm
Vias	82 × Ø0.30 mm drill / Ø0.60 mm pad
Annular ring	0.15 mm
Through-hole parts	None

Drill	Qty	Use
Ø0.30 mm PTH	82	Vias
NPTH	0	—
Total holes	82	

04 ASSEMBLY

Technology	Double-sided SMT; no through-hole parts
Component side	27 parts: 2 × ADS131M08 (5 × 5 mm, 32-pin), 8.192 MHz oscillator, on-board low-noise supplies, passives (0402/0603), 2 × XIAO sockets, stacking connector
Pin side	18 pogo pins (Ø4 × 11 mm, SMD-mounted)
Totals	45 placements · 28 unique parts · 286 SMD pads
Stencil	Both sides (component side fine-pitch 0.5 mm; pin side pogo pads)
Pogo pins	Placed and soldered square to the board; check that all 18 move freely after reflow
Jumpers	0 Ω 0402 links set in production for single-board use (see OC16-PIN-004)

05 PRODUCTION QUALITY TEST

1 Visual
All 18 pogo pins upright, same height and springing back freely; no solder on the plungers.

2 Firmware
Fit a XIAO ESP32-S3 (5V / GND / 3V3 aligned), flash the official firmware; it must advertise as Octopus-16-XXXX.

3 ADC detect
Connect from PiEEG Cloud or the Python SDK. The LED must stay solid (no fast double-blink) and all 16 channels must stream at 250 SPS with a continuous counter.

06 HANDLING AND CARE

Topic	Requirement
ESD / storage	Handle by the edge on an anti-static surface; store dry with the pins protected from side loads.
Pogo pins	Press straight down only. Clean with isopropyl alcohol > 90 %; never oil or grease.
Classification	Research and education tool. Not a medical device.
Design files	Circuit, layout and parts list are not included in the user documentation.

07 RELATED DOCUMENTS

OC16-DS-001
Product Datasheet

OC16-MD-002
Mechanical & Dimensional Drawing

OC16-PIN-004
Electrode Array & Pinout Guide

OC16-QSG-005
Quick Start & Safety Guide

OC16-SW-006
Bluetooth Interface & Software Guide